



FACULTY OF COMPUTING

SEMESTER 2/20252026

SECI1113 – SYSTEM ANALYSIS AND DESIGN

SECTION 01

PROJECT PHASE 2

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1.0 OVERVIEW OF THE OBJECT

FlexHub is a marketplace platform that focuses on rental properties and event spaces. In this project, the selected theme is Theme A :FlexRent (Property Super-App), which mainly involves managing different types of rentals including hourly even spaces, short-term stays and long term rentals. The purpose of the platform is to connect customers with property providers in a more convenient and manage their rental services through the platform.

However the current FlexHub system still operates manually using third-party messaging. Most communication between customers and providers is done through WhatsApp or other messaging application. Providers then manually check their schedules via their notebooks or personal calendar before responding to the customers. Payments are also handled manually through online banking transfers, where customers must send their payment receipts to providers for verification.

Because the system depends heavily on manual work, several issues occur in the booking and management process. The current workflow is slow, inefficient and difficult to manage especially when handling many bookings at the same time. There is also a higher risk of human errors such as double bookings, delayed responses, missing payment records and incorrect scheduling. These problems can negatively affect customer's satisfaction and reduce the efficiency of the business operations.

Therefore, this project is carried out to analyze the current AS-IS System and identify the weaknesses in the existing workflow. Based on the analysis, a new TO-BE System will be proposed to improve the overall operation of FlexRent. The proposed system aims to automate important processes such as booking management, payment verification, scheduling updating and cancellation handling. By implementing a more centralized and automated platform, the system can provide a smoother, faster and more reliable experience for both users and providers. In addition, the proposed solution can help reduce workload, improve efficiency and support better data management within the platform.

2.0 PROBLEM STATEMENT

The current FlexHub system currently operates using a highly manual workflow that depends heavily on third-party messaging platforms such as WhatsApp for communication, booking coordination, and payment confirmation. Most interactions between users and providers are conducted through WhatsApp, an app for communication. This manual process creates operational inefficiencies, delays and a high risk of human error, affecting both users and service providers.

One of the major problems identified in the current “AS-IS” system is the lack of an automated booking management system. When users asked about available booking slots, providers need to manually check their physical notebooks or personal calendars before responding. This causes slow response time, double-booking risks and scheduling conflicts, especially when handling multiple customers simultaneously.

Besides that, payment processing is also performed manually. Providers need to send their bank account details through WhatsApp, while users must transfer the payment and send the receipt as proof. Provider then manually verify transactions through their banking portals before confirming bookings. This process is time-consuming, inefficient and vulnerable to fraud or missing payment records.

The current system also lacks an efficient booking modification feature. Users who wish to reschedule or modify their bookings must contact providers directly through chat. Providers are then required to manually update schedules and rearrange booking slots. This may lead to inaccurate booking records.

Furthermore, cancellation and refund handling are entirely manual. Providers must manually calculate refund amounts based on their own policies, collect customer banking information and manually transfer refund payments. This process can delay refund handling and may result in calculation errors or customer dissatisfaction.

Overall, the dependency on manual processes reduces operational efficiency, limits scalability and negatively impacts user experience. Therefore, an automated AI enhanced system is needed to optimize booking operations, payment validation, schedule modification processes and refund management while enhancing system accuracy, operational efficiency and overall customer satisfaction.

3.0 PROPOSED SOLUTIONS

To address the challenges identified in the problem statement, this project proposes the “TO-BE” system is designed to replace the current manual workflow with a fully integrated web and mobile platform that automates booking management, payment processing, customer communication, and scheduling processes. By eliminating the dependency on WhatsApp communication, the system can improve operational efficiency, reduce human errors, and provide a more organized booking experience

The proposed platform will include several automated features such as real-time booking availability, integrated online payment gateways, automated notifications, and self-service dashboards. Users will be able to check available booking slots instantly without waiting for manual responses from providers. Once a booking is confirmed, the system will automatically update schedules and send booking confirmations, payment updates, and reminder notifications to users and providers.

In addition, the payment process will be fully digitalized through a secure payment gateway integrated into the platform. Users can make payments directly through the system, while providers can receive automatic payment verification without manually checking bank transfers or payment receipts. This feature helps reduce transaction processing time and minimizes the risk of payment fraud or missing records.

The proposed system will also support automated booking modification and cancellation management. Users can reschedule or cancel bookings directly through the platform, while the system automatically updates schedules and manages slot availability. Refund calculations will also be processed automatically based on predefined cancellation policies, reducing manual workload and improving accuracy.

The proposed system will also integrate the AI Dynamic Pricing feature to make price management more efficient and flexible. In the current system, pricing is handled manually, which can lead to inconsistent prices and difficulties in adjusting prices based on customer demand. With the AI Dynamic Pricing feature, the system can automatically adjust rental or service prices in real time according to factors such as booking demand, seasonal trends, service availability, and booking frequency. This allows service providers to manage pricing more easily without needing to manually update prices all the time. At the same time, customers can receive fair and updated pricing information based on current demand. By automating the pricing process, the system can help reduce workload, improve operational efficiency, increase booking opportunities, and provide a better overall experience for both customers and service providers.

In addition, the proposed system will implement the AI Smart Match feature to provide personalized recommendations for users. The system analyzes user preferences booking history, selected categories and uploaded reference images to suggest suitable dresses or costumes automatically. This helps users find rental items more quickly and improves their overall booking experience. At the same time, providers can increase booking opportunities because their listings are matched with relevant customers. By integrating AI Smart Match, the FlexRent system becomes more efficient, user-friendly and intelligent for both customers and providers.

4.0 INFORMATION GATHERING PROCESS

To successfully transition FlexRent “AS-IS” from its current state, a highly manual and static directory into the FlexRent “TO-BE” platform, a detailed requirements gathering process was essential to accurately define the system's core features. Understanding user behaviour across mixed time intervals (hourly event bookings, daily short term stays and long term residential) required comprehensive insight from target stakeholders. The primary goal of this phase was to evaluate the AS-IS system and define functional specifications for the automated, AI-enhanced TO-BE system and to identify the stakeholder's current operational issues and gather clear requirements for the new platform.

4.1 Method Used

The selection of an unobtrusive method for gathering requirements was based on its ability to collect data without directly interfering with the stakeholder. For this phase, a digital questionnaire was designed and distributed to the stakeholder using Google Forms. This approach proved highly effective for the system study for several key reasons.

This method was ideal for obtaining honest and natural responses. When an individual is interviewed in person or observed while working, they may feel pressured or uncomfortable when answering questions. By utilizing a private online form, the stakeholder was able to comfortably share real experiences, opinions, and complaints regarding the current manual property directory system without feeling any external pressure.

In addition, using a digital survey through Google Forms helped avoid interrupting the stakeholder's busy schedule. The stakeholder could access the form and answer the questions at a convenient time, eliminating the need to arrange live meetings or disrupt daily work routines. This method also enabled the project team to gather both structured factual information and detailed personal opinions simultaneously.

Before distributing the survey, several corrections and improvements were made to ensure the clarity and effectiveness of the questionnaire. On 7 May 2026, corrections and refinements were conducted on the Google Form to improve the wording, structure, and clarity of the questions. Afterwards, on 14 May 2026, the final corrections and reviews were completed before the survey was officially distributed to the stakeholder.

The Google Form consisted of a total of 19 questions, which included 15 close-ended questions and 4 open-ended questions. The questionnaire was organized into several sections to ensure a systematic and user-friendly flow for the stakeholder. The sections were structured as follows:

1. User Role / Identity – Questions related to the stakeholder's role and background.
2. Current System (AS-IS) – Questions focusing on the existing manual property directory system and its current issues.
3. Proposed System Requirements (TO-BE) – Questions regarding the expectations and requirements for the proposed system.
4. AI Feature – Questions related to the implementation and usefulness of AI-based features in the proposed system.
5. Booking System – Questions regarding booking and reservation functionalities.
6. Payment System – Questions focusing on payment methods and transaction features.
7. Additional Requirements – One open-ended question allowing the stakeholder to provide additional suggestions, comments, or recommendations for the system.

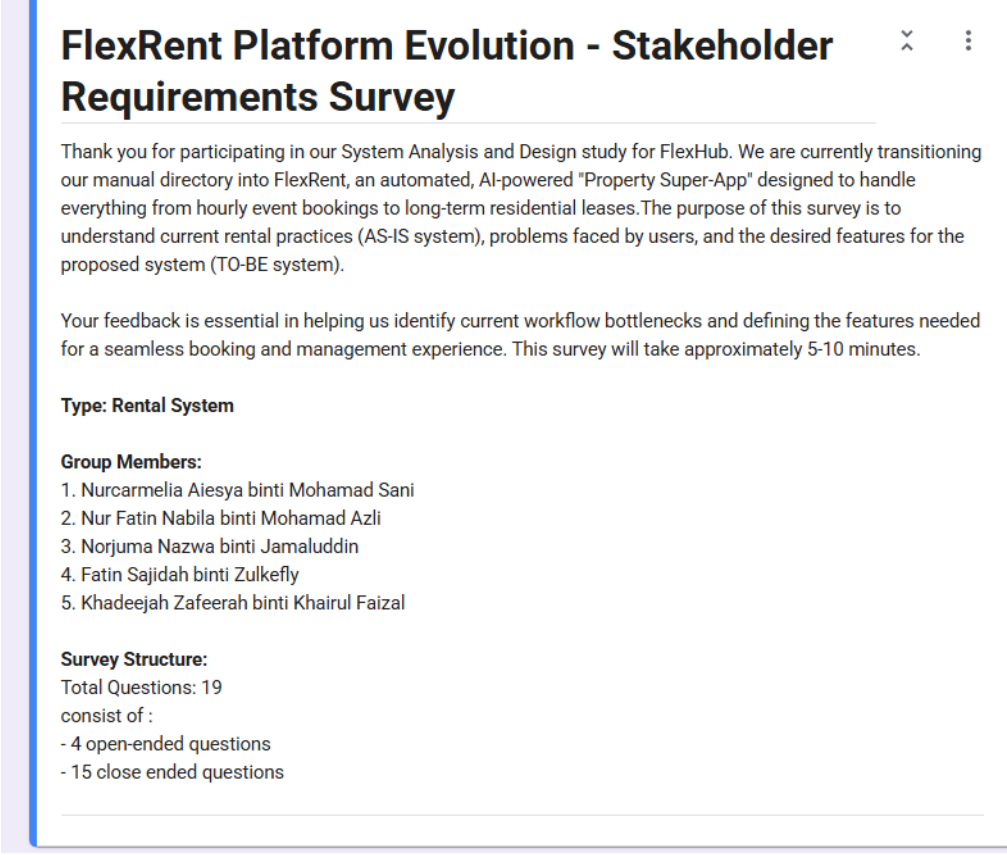
Overall, the use of Google Forms as an unobtrusive requirement gathering method provided flexibility, convenience, and efficient data collection while ensuring the stakeholder could provide honest and meaningful feedback for the development of the proposed system.

4.2 Summary from method used

The information gathering for the FlexRent platform evaluation was successfully conducted by using an online Google Form questionnaire as an unobtrusive method to collect system requirements directly from the primary stakeholder. This approach was highly effective because it allowed the stakeholder to comfortably share an honest feedback and genuine complaints about the current manual property without feeling pressure in face to face interview while also preventing any disruption to their busy daily routine managing mixed interval listings.

The survey was systematically structured with a total 19 questions, consisting of 15 close ended questions and 4 open ended questions to gather detailed, written description of specific workflow and desired features. Ultimately, the data collected provides clear baseline of the current AS-IS system problems and establishes the necessary user requirements that will use to build the system models including Data Flow Diagrams (DFDs) and Entity Relationship Diagrams (ERDs) for the new TO-Be platform.

Example of Survey



FlexRent Platform Evolution - Stakeholder Requirements Survey

Thank you for participating in our System Analysis and Design study for FlexHub. We are currently transitioning our manual directory into FlexRent, an automated, AI-powered "Property Super-App" designed to handle everything from hourly event bookings to long-term residential leases. The purpose of this survey is to understand current rental practices (AS-IS system), problems faced by users, and the desired features for the proposed system (TO-BE system).

Your feedback is essential in helping us identify current workflow bottlenecks and defining the features needed for a seamless booking and management experience. This survey will take approximately 5-10 minutes.

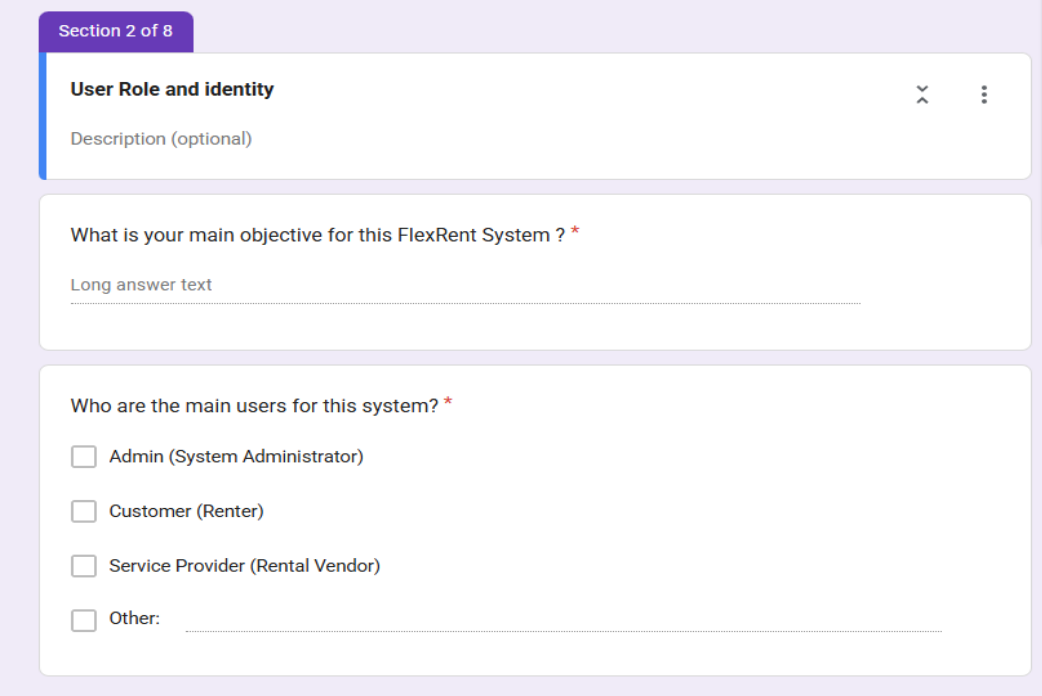
Type: Rental System

Group Members:

1. Nurcarmelia Aiesya binti Mohamad Sani
2. Nur Fatin Nabila binti Mohamad Azli
3. Norjuma Nazwa binti Jamaluddin
4. Fatin Sajidah binti Zulkefly
5. Khadeejah Zafeerah binti Khairul Faizal

Survey Structure:
Total Questions: 19
consist of :
- 4 open-ended questions
- 15 close ended questions

4.2(a) shows the Google Form cover page



Section 2 of 8

User Role and identity

Description (optional)

What is your main objective for this FlexRent System ? *

Long answer text

Who are the main users for this system? *

☐ Admin (System Administrator)

☐ Customer (Renter)

☐ Service Provider (Rental Vendor)

☐ Other: _____

4.2(b) shows the section of user role and identity

Current System (AS-IS)

Description (optional)

How do you currently rent a dress or costume ? *

☐ Physical walk-in shops
 ☐ Personal contacts
 ☐ Social Media (Instagram/TikTok/Facebook)
 ☐ Existing general marketplaces (Carousell/Mudah.my)

What is the biggest problem you face with the current rental system *

Long answer text

How are rental agreements and deposits currently recorded? *

☐ Manual logbooks/Paper receipts
 ☐ Personal spreadsheets (Excel/Google Sheets)
 ☐ No formal record, just chat history
 ☐ Physical cash deposit only
 ☐ Other:

On a scale 1-5, how secure do you feel about the current payment and return process? *

1-Not Secure
 2-Somewhat Secure
 3-Neutral
 4-Secure
 5-Very Secure

1	2	3	4	5
☆	☆	☆	☆	☆

4.2(c) shows the section for Current System (AS-IS)

Proposed System Requirement (TO-BE)

Description (optional)

Admin View

This is designed for the system administrator to manage and control the overall rental platform

Function of Admin :

- Manage user accounts (customers and renter)
- Approve or reject renter registrations
- Monitor all bookings and payments
- Handle refund request or report from users
- Manage costume categories, pricing and availability
- Ensure smooth system operation and data management

Are these Admin features sufficient ?

☐ Yes, All are suitable
☐ No, need to add something (click other)
☐ Other:

Customer View

This is designed for users to browse and rent costumes and dresses.

Function of Customer :

- Register and log in to system
- Search and filter costume/dresses
- View product details (size, price, availability/ratings)
- Book or rent costumes based on availability
- Make online payments
- Track rental status and rental dates
- Leave reviews and ratings after rental

☐ Yes, All are suitable
☐ No, need to add something (click other)
☐ Other:

Service Provider (Rental Vendor) View

This is designed for costume or dress rental providers to manage their rental items and customer bookings

Function of Vendors

- Register and manage shop profile
- Upload costume or dress listings
- Update item availability, sizes and pricing
- Accept or reject rental request
- Monitor booking payments
- Manage customer communication and rental returns

☐ Yes, All are suitable
☐ No, need to add something (click other)
☐ Other:

What core features should the system include ?

- ☐ Costume & Dress Search Filter
- ☐ Rental Booking System
- ☐ Online Payment System
- ☐ Reviews & Ratings
- ☐ Profile Management
- ☐ Rental Tracking & Return Reminder
- ☐ Wishlist / Favorite Items
- ☐ Size & Availability Checking
- ☐ Other:

4.2 (d) shows the section for proposed system requirements (TO-BE)

AI Features

Description (optional)

What AI features should be included ? *

- ☐ Smart Recommendation (Recommends the most suitable costumes or dresses to users based on prefer...
- ☐ AI Size Recommendation (Suggests the most suitable size for users based on body measurements or pr...
- ☐ Dynamic Pricing (Adjusts rental prices automatically based on demand, rental duration, festive seasons, ...
- ☐ AI Trend Suggestion (Suggests trending or popular costumes based on current events, themes, or seaso...
- ☐ Smart Availability Scheduler (Manages rental dates, prevents double bookings, and tracks return schedul...
- ☐ AI Chatbot Support (Provides instant customer support and answers common rental questions.)
- ☐ Other:

How should the system handle the new renter/vendors with no ratings? *

Long answer text

4.2 (e) shows the section for AI features

Booking System

Description (optional)

Should buffer time be included between rentals? *

Example :

Rental 1 : 10:00 AM - 2:00 PM

Cleaning & preparation Buffer : 2:00 PM - 4:00 PM

☐ Yes
 ☐ No
 ☐ Other:

How long should the cleaning/preparations buffer time be ? *

☐ 1 hour
☐ 2 hours
☐ Depends on the dress/costume type
☐ Vendors can set manually
☐ Other:

Should user be able to cancel or reschedule bookings? *

☐ Yes
 ☐ No

4.2 (f) shows section for booking system

Payment System

Description (optional)

What type of payment system should be allowed *

☐ Full Payment
☐ Deposit Payment
☐ Installment Payment
☐ Other:

Which payment should be included ?

☐ FPK (Online Banking)
☐ Credit/Debit Card
☐ E-Wallet (Touch 'n Go, GrabPay, Boost)
☐ Cash payment
☐ Other:

Should the system include a security deposit for rented costumes/dresses? *

☐ Yes
☐ No
☐ Depend on costume/dress type
☐ Other:

4.2 (g) shows section for payment system

Additional Requirements

Description (optional)

Any additional features or expectations for the development of FlexRent System ? *

Long answer text

4.2 (h) shows section for additional requirements

5.0 REQUIREMENT ANALYSIS

5.1 CURRENT BUSSINESS PROCESS

The current business process of FlexRent is still highly manual and depends heavily on communication through WhatsApp. Most activities such as booking inquiries, payment verification, booking modification and cancellation handling are performed manually by the property providers. This workflow causes delays, inefficiencies and increases the possibility of human errors.

The process begins when customers browse available rental properties or event spaces on the FlexHub platform. After finding a suitable property, customers will click the “Contact via WhatsApp” option to communicate directly with the provider. Customers then ask about the availability of the property, booking date and time, rental price and other related information details.

After receiving the booking inquiries, the provider manually checks their schedule using their notebooks or personal calendar to confirm whether the requested date and time are available. Once the provider confirm the availability, both customer and provider proceeds with the booking discussion through the WhatsApp application.

Next, the payment process is carried out manually. The provider sends their bank account details to the customer, and the customer transfers the payment through online banking. After completing the transaction, the customer sends a screenshot of the payment receipt to the provider as proof of payment.

The provider then manually log into their banking system to verify whether the payment has been succesfully received. Once the payment is confirmed, the provider manually updates the booking schedule and sends a booking confirmation message to the customer through WhatsApp.

If the customer wants to modify the booking date or time, they need to. Ontact the provider again through WhatsApp. The provider manually, checks the schedule, removes the previous booking slot and updates the new booking information. This process may create scheduling conflicts if the records are not updated properly.

For cancellations, customers also need to message the provider directly. The provider manually calculates the refund amount based on their own cancellation policy. After that, the provider transfers the refund manually back to the customer’s back account and updates the booking record themselves.

Overall, the current AS-IS workflow is inefficient because all processes depend on manual communication, manual payment verification and manual schedule management. The lack of automation and centralized data management increases workload, slows and dwn operations and the overall customer experience.

5.2 FUNCTIONAL REQUIREMENT

5.2.1 Manual Booking Slot Checking

The manual booking slot checking process allows users to contact the service provider through WhatsApp to ask about available booking dates and times. The provider then checks availability using a notebook or calendar manually to verify slot availability time slots before responding to the user. After checking the schedule, provider will send the availability slots to the customer through the chat. Once the user selects a preferred date and time, the provider manually confirm and records the booking in their schedule.

5.2.2 Manual Payments Via WhatsApp

The manual payments via WhatsApp allows providers to send their bank account details to users through WhatsApp for payment purposes. Users can perform online bank transfers using their preferred banking platform and send a screenshot of the payment receipt as proof of transaction. Providers then manually log into their bank portals to verify whether the payment has been successfully received. After confirming the payment, the provider manually updates the booking status and informs the user that the booking has been confirmed.

5.2.3 Manual Booking Modifications

The manual booking modifications allows users to request booking schedule changes through WhatsApp chat with the provider. After receiving the request, the provider manually checks for available replacement time slots in their notebook or calendar. Once a suitable slot is found, the provider removes the previous booking entry and updates the schedule with the new selected date and time. Finally, the provider confirms the modified booking details with the user through messaging manually.

5.2.4 Manual Cancellations & Refund

The manual cancellations and refund allows users to contact providers through messaging applications like WhatsApp to request booking cancellations. Providers manually review the cancellation request and calculate the refund amount based on their own cancellation and penalty policies. Users are required to provide their bank account details so the refund can be transferred back to them. After calculating the refund, the provider manually performs the bank transfer and informs the user once the refund process has been complicated

5.3 NON-FUNCTIONAL REQUIREMENTS

5.3.1 Manual Booking Slot Checking

The system's operational quality and process conditions are defined by several key non-functional requirements. To ensure high availability and seamless usability, providers must be consistently reachable through WhatsApp, which means providers should reply within a reasonable time. System reliability and accuracy are critical to core operations. The system must strictly prevent double bookings, and displayed booking availability must always perfectly match the provider's real-time schedule. Finally, communication should be easy for both users and providers.

5.3.2 Manual Payments Via WhatsApp

The system's payment processing and data handling are controlled by strict operational quality standards. To guarantee strong security, so all bank and payment details will be protected. Next for traceability, all payment records and receipts must be fully trackable for auditing purposes. To make sure system accuracy and reliability, the payment verification process must be entirely correct and payment confirmations must be safe to prevent transaction drops. In addition, the entire verification process must execute reasonably fast, ensuring a smooth and efficient user experience.

5.3.3 Manual Booking Modifications

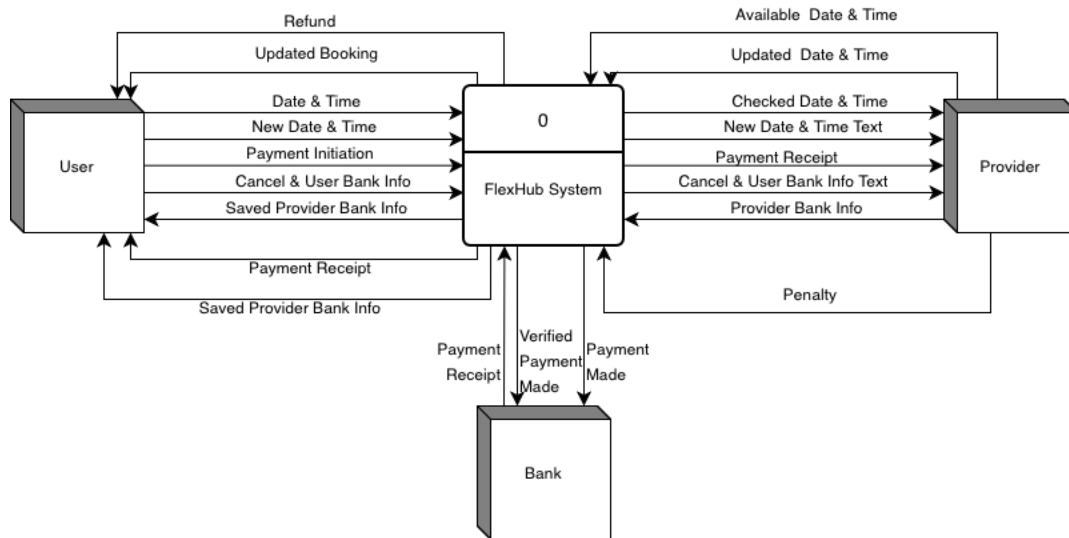
The system's manual booking modifications must meet certain criteria to ensure an efficient operation. The schedule updates for the booking must remain correct in order to keep the consistency of the system. The system should also never risk overwriting other active bookings to make sure the reliability of the system. To minimize any delays, the system must deliver a fast response time by processing modification requests quickly. Lastly, the system must be able to accommodate multiple schedule changes at once and keep the booking records simple and straightforward to maintain flexibility and maintainability.

5.3.4 Manual Cancellation and Refunds

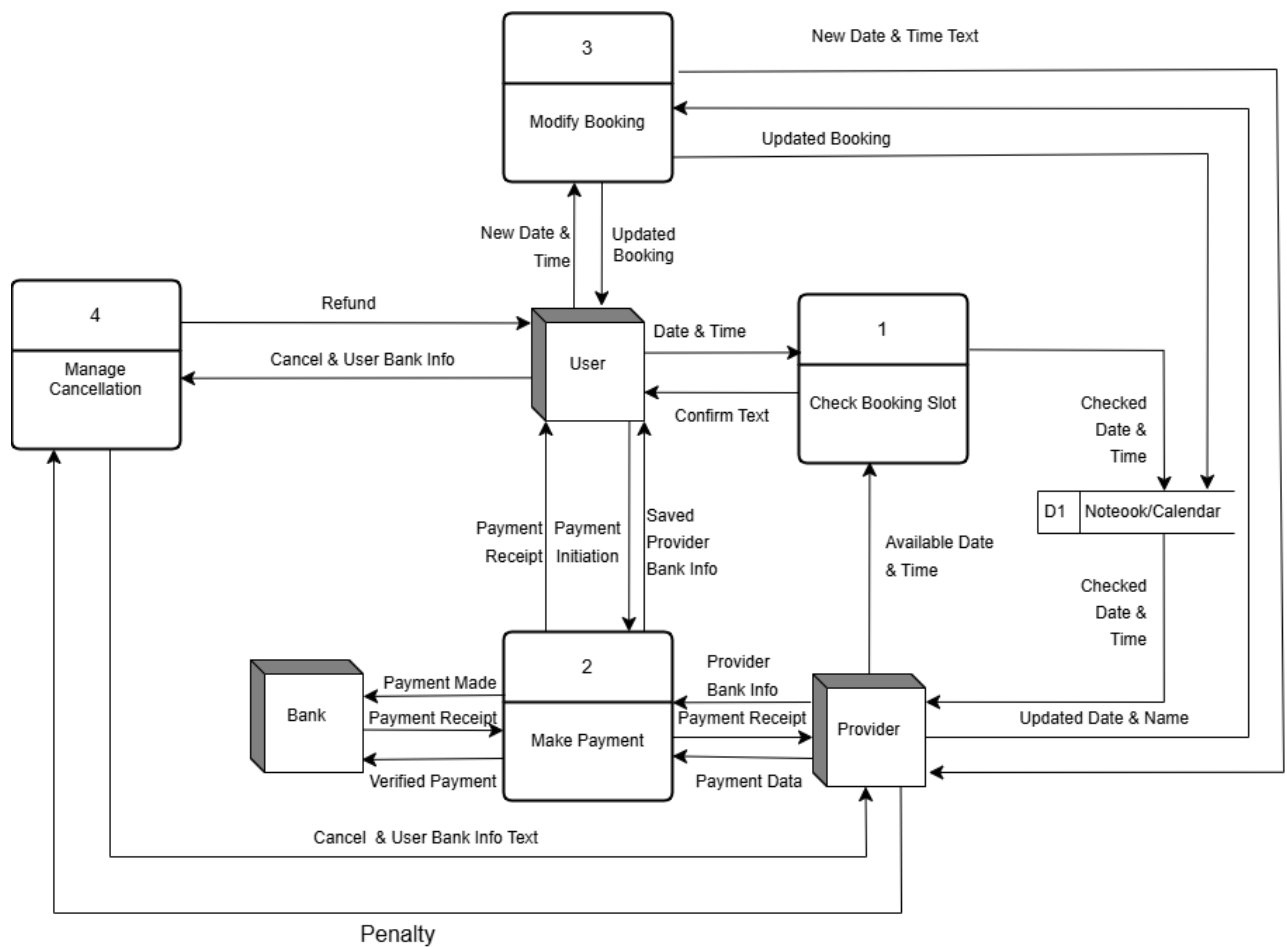
The system's cancellation and refunds must meet specific performance and safety benchmarks to make sure a smooth operation. To ensure maximum security and auditability, all user banking information must remain completely confidential and all of the refund actions must be recorded for reference. Next, the refund calculation must follow policy correctly and the transactions should be completed successfully to protect the financial integrity. Finally, to maintain a high standard of service, the workflow emphasizes efficiency which requires refund requests to be processed quickly and without any lag.

5.4 Logical DFD AS-IS System (Context Diagram, Diagram 0, Child Diagram)

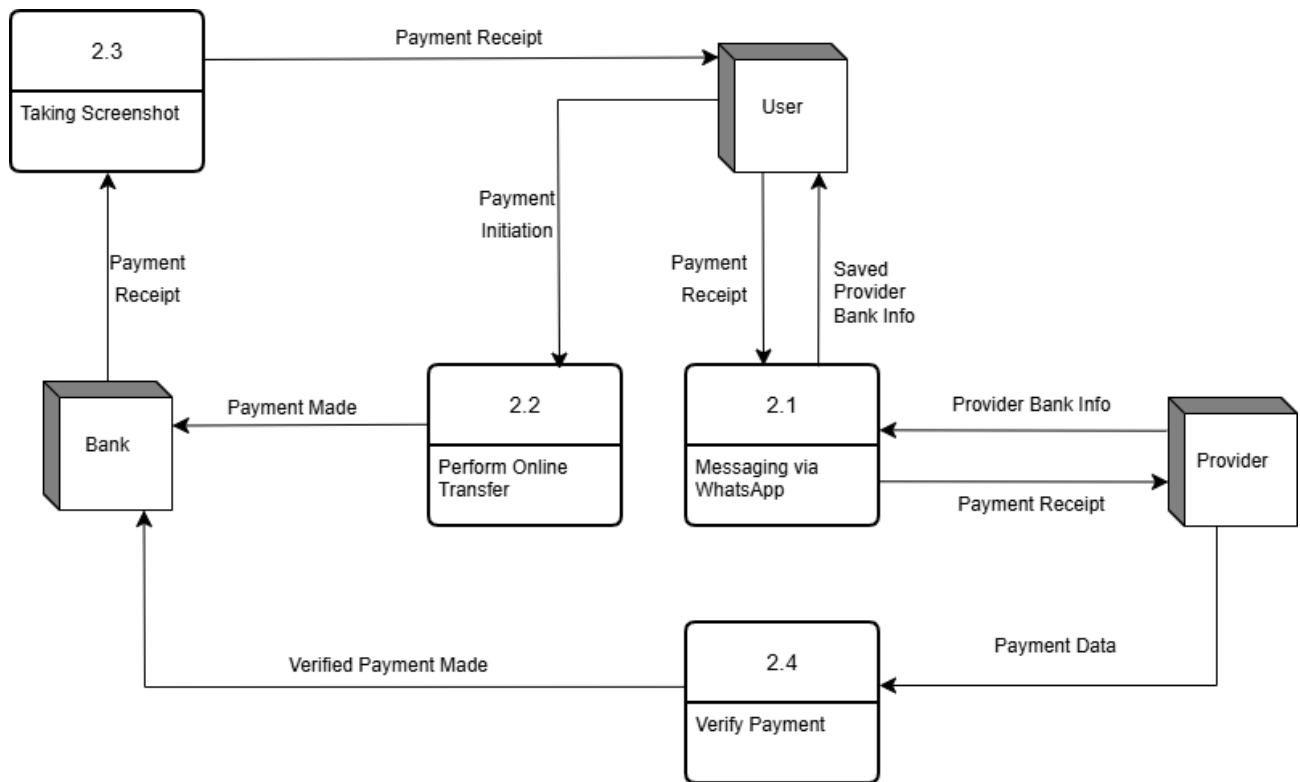
5.4.1 Context Diagram



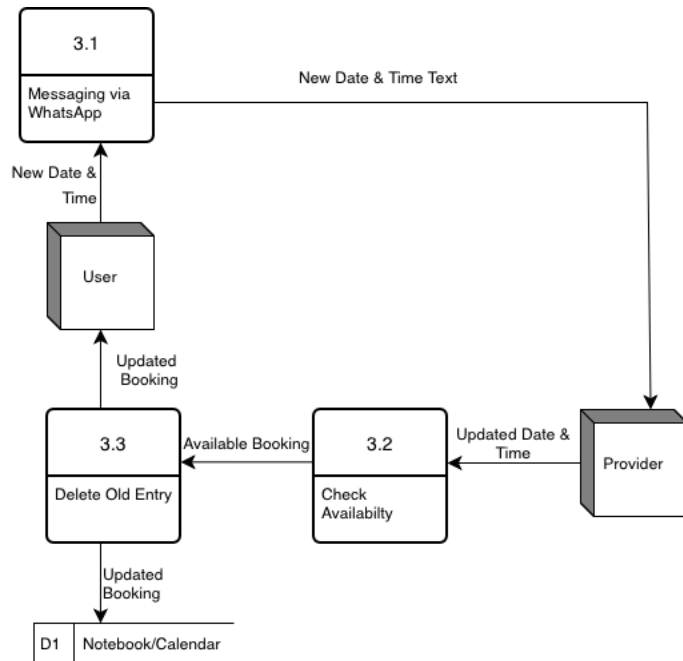
5.4.2 Diagram 0



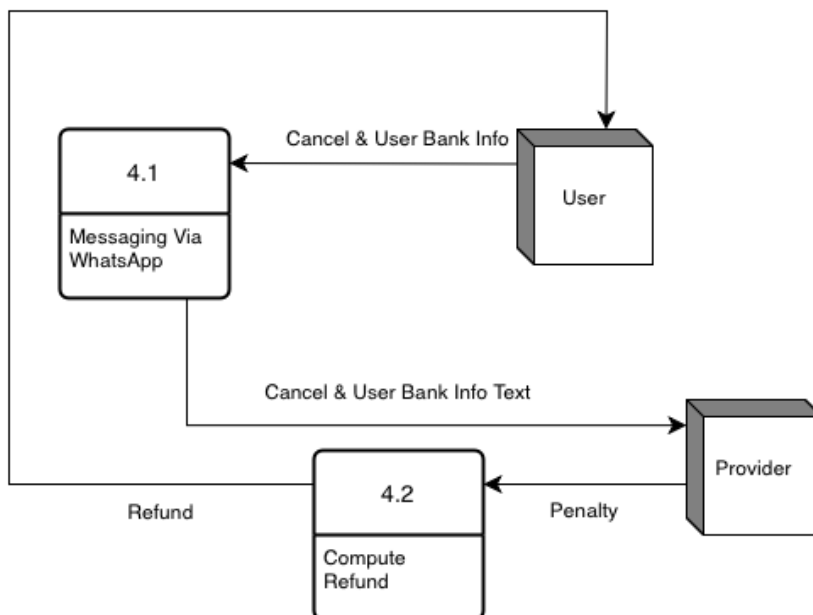
5.4.3 Manual Payment Via WhatsApp Child Diagram



5.4.4 Manual Booking Modifications Child Diagram



5.4.5 Manual Cancellations & Refunds Child Diagram



6.0 SUMMARY OF REQUIREMENT ANALYSIS PROCESS

The requirement analysis process for the FlexHub AI-Powered Marketplace Migration (Flexrent) project successfully transitioned the system from a high manual static directory into a structured, fully automated 'TO-BE' design framework. This process was executed through a systematic evaluation of stakeholder needs, operational bottlenecks and rigorous system modelling.

1. Information Gathering

The process initiated with an investigation into the current 'AS-IS' operational workflows. Through interactive and unobtrusive information-gathering methods, the critical operational inefficiencies of the legacy system were mapped out. The analysis targeted four primary manual dependencies that currently rely entirely on third-party messaging and cross-referencing:

- **Manual Booking Slot Checking:** Providers cross-referencing physical notebooks or personal digital calendars to confirm availability.
- **Manual Payments:** Offline banking transfers verified via screenshot receipts and manual banking portal logins.
- **Manual Booking Modifications:** Manual erasure and blocking of calendar entries through text chats.
- **Manual Cancellations and Refunds:** Manual computation of custom penalty rules and manual return transfers.

2. Specification of Functional and Non-Functional requirements

Based on the identified bottlenecks, user and provider requirements for the automated 'TO-BE' platform were derived and formalized into a Requirements Specification Document that incorporates a multi-layered artificial Intelligence architecture:

- **Functional Requirements:** Formulated to eliminate manual workflows by incorporating real-time calendar availability, self-service dashboards and an integrated payment gateway. To meet the advanced system criteria, two distinct AI capabilities were woven into the core architectural design. First, an AI Smart Match engine analyses historical seeker behaviour, sizing preferences and style aesthetics to automatically generate tailored costume recommendations. Second, an AI Dynamic Pricing engine acts as the primary operational optimization framework, evaluating seasonal high-demand periods and platform apparel scarcity to automatically adjust rental rates dynamically across hourly, daily and weekly lease intervals.
- **Non-Functional Requirement:** Defined strict parameters ensuring system security, payment gateway integrity, data processing privacy for user preference profiling and specific real-time calculation latency performance metrics for both AI execution paths.

3. Logical Modelling (AS-IS DFD)

To visualize and align the data flows within the manual system, a complete Logical 'AS-IS' Data Flow Diagram package was developed. This modelling consisted of a Context Diagram defining the boundary of the current manual system and its external entities (Seekers and Providers), alongside a Diagram 0 decomposing the core manual processes to accurately map the inputs, processes, outputs and data stores driving the manual WhatsApp flows.